

ZOO4307C Lecture 2

2026. 8. 25

Notes compiled by ZX

Systematic phylogenies:

- a. **Sympomomorphy**→ shared because it was derived in their most recent common ancestor
- b. **Symplesiomomorphy**→ retained ancestral trait, inherited from a common ancestor but **does not mean close evolutionary relationship**
- c. **Homoplasy**→ shared because the same trait originated **independently more than once**→ bird/bat wings

Common sources: parallel evolution, convergent evolution

- d. **Autapomorphy**→ not shared, derived in a single lineage

Define mono, para, polyphyletic groups

- a. **Monophyletic:** AKA clade, common ancestor + all of its descendants
- b. **Paraphyletic:** common ancestor, but does not include all descendants
- c. **Polyphyletic:** Grouping lineages that are not closely related based on superficial similarities.

*Fossils help people to put actual dates on phylogenies.

Geologic time scale: eon > era > period

*We focus ONLY on phanerozoic EON.

1. **Paleozoic era:** cambrian, ordovician, silurian, devonian, carboniferous, permian
 - ❖ Cambrian: 541, aquatic, chordates already present, div of marine orgs
 - ❖ Ordovician: 485, div of marine orgs, seeing some of the first vert, orgs getting larger
 - ❖ Silurian: 444, plants and arthropods move to land, verts still in water
 - ❖ Devonian: 419, age of fishes, origin of tetrapods, first verts on land (first major terre transition)
 - ❖ Carboniferous: 360, origin of amniotes, huge tropical swamps (today's fossil fuel), large amphibians
 - ❖ Permian: 300, terre vertes div, NO modern mammalia, ancestors of reptiles/mammals div
 - ❖ End-Permian event: 252, most severe mass extinction, 96 multisel- went extinct, massive volcanic activity→ greenhouse gases, climate/sea level change, Siberian Traps
2. **Mesozoic era:** triassic, jurassic, cretaceous, K-Pg Mass Extinction
 - ❖ Triassic: 252-201, div of reptiles and earliest mammals, filling those empty niches, very early dinosaurs appeared in this period, adaptive radiation, Pangea→Lauraria + Gondwana
 - ❖ Jurassic: 201-145, div of dinosaurs, first pterosaurs, birds, and lizards

- ❖ Cretaceous: 145-66, div of dinos, first snakes, most dinos are here
- ❖ K-Pg Mass Extinction: 66, asteroid impact, Yucatan peninsula, non-avian dinosaurs extinct, large terrestrial spp most affected, 76% spp wiped out

3. Cenozoic era: tertiary, quaternary

- ❖ Tertiary: 66-2.6, div of most recognizable modern terre vertebrates
- ❖ Quaternary: 2.6-now, ice ages, mammalian megafauna, and hominids

Vertebrates: where do they come from?

*bilateria-deuterostomia-chordata-vertebrata

1. Phylogenetic tree:

2. Bilateria: bilateral symmetry, triploblasty, tube-in-tube body plan, and gastrulation

- a. Bilateral symmetry
- b. Triploblasty: ectoderm, mesoderm, and endoderm
 - ❖ Ectoderm:
 - ❖ Mesoderm:
 - ❖ Endoderm:
- c. Body plan: tube-within-a-tube, outer body wall + digestive tract
- d. Gastrulation:

3. Protosome vs. deuterostome

- ❖ Protostome: blastopore → mouth, second opening → anus
- ❖ Deuterostome: blastopore → anus, second opening → mouth

4. Chordata: cephalochordata, urochordata, and vertebrata

*Urochordates are closer to vertebrates than cephalochordates based on DNA.

- ❖ Cephalochordata:
- ❖ Urochordata/ tunicata:
- ❖ Vertebrata:

5. FIVE diagnostic features of chordates: notochord, dorsal nerve chord, myomeres, post-anal tail, and endostyle (thyroid homolog)

- ❖ Notochord:
- ❖ Dorsal nerve chord:
- ❖ Myomeres:
- ❖ Post-anal tail:
- ❖ Endostyle/ thyroid homolog:

6. Non-chordate deuterostomes: hemichordates and echinoderms

*Hemichordates & chordates: similarity does not automatically mean homology → homoplasy

- ❖ Hemichordates:
- ❖ Echinoderm: